# AI-Powered Personal Health Companion for Early Risk Detection

**Problem Statement Title -** A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.

**Theme -** MedTech / BioTech / HealthTech

**Category –** Hardware

**Team Name –** STRAW HATS

# PROPOSED SOLUTION

**AERIS: AI-BASED EARLY RISK & INTELLIGENT SAFETY COMPANION**

## Problem Vs Our Solution

| Existing Problem | AERIS Solution |
|---|---|
| Single-sensor monitoring | Multi-sensor risk detection |
| Fixed health thresholds | Personal baseline |
| Internet dependency | Online + Offline LoRa |
| Delayed alerts | Early risk detection |
| No environmental context | Health + environment correlation |
| Raw data monitoring | **0–100 Risk Index** |
| Limited personalization | Personalized guidance |
| Cloud-dependent processing | Edge-first, privacy-focused processing |

## Our Solution




## Why We Stand Out

- **Multi-Modal Risk Correlation :** Correlates **physiological + mobility + environmental signals** instead of interpreting isolated sensor values.
- Converts **interacting signal changes** into one interpretable **0–100 AERIS Risk Index** instead of isolated sensor alerts
- **RISK-TO-ACTION, NOT JUST MONITORING**
  Converts detected patterns into a **0–100 AERIS Risk Index → guidance → alerts → escalation**.

STRAW HATS
TEAM STRAW HATS

# TECHNICAL APPROACH

**"AERIS detects the direction of risk, not just the current sensor value."**

## 01 SENSING
Collect real-time data

- Heart Rate / HRV
- $SpO_2$
- Body Temperature
- Motion / Activity
- Stability / Gait
- Environment (Temperature, Humidity, AQI)



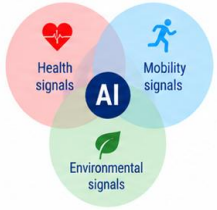
Multiple signals.
A complete picture.

## 02 EDGE PROCESSING
On-device intelligence (ESP32)



- Signal Filtering
- Noise Removal
- Feature Extraction
- Personal Baseline
- Local Analysis (TinyML)
- Privacy-First Processing on Device

## 03 MULTI-MODAL AI
Correlate & detect patterns




- Pattern Detection
- Anomaly Detection
- Context-Aware Risk Evaluation

## 04 AERIS RISK INDEX
Unified risk score (0 – 100)




Personalised | Real-time
Context-aware

## 05A ONLINE INTELLIGENCE
Deeper analysis & personalised guidance

ESP32 (Internet) → AI + Weather API → Mobile App / Web Dashboard → Guidance + Notifications

## 05B OFFLINE SAFETY
Works without internet (LoRa)

LoRa Sender → LoRa (Link Long Range) → LoRa Receiver → Buzzer / LED / Vibration

User can turn ON the buzzer if uncomfortable (if safe). → If risk is detected, the buzzer turns ON automatically. → If the buzzer is not turned OFF within 8 seconds (after manual or auto trigger). → Emergency alert will be sent automatically (LoRa + Notification).

## 06 ACTION – From risk to real-world response

| 0 – 25 LOW RISK | 26 – 50 MODERATE RISK | 51 – 75 HIGH RISK | 76 – 100 CRITICAL RISK |
|---|---|---|---|
| Continue Monitoring | Guidance & Precautions | Alert User & Caregiver | Emergency Alert (LoRa + Notification) |

**A HEALTHIER, SAFER TOMORROW**
FOR INDIVIDUALS, FAMILIES AND COMMUNITIES

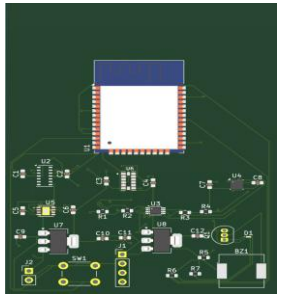
## EDITABLE SYSTEM ARCHITECTURE & OFFLINE COMMUNICATION FLOW

1. **Sensors** — Data Acquisition (PPG, Temp, IMU, Env)
2. **ESP32 Core** — Local Processing & Noise Filtering
3. **Edge AI Engine** — TinyML Anomaly & Feature Extraction
4. **Risk Index** — 0–100 Multi-Modal Unified Scoring
5. **Alert System** — Threshold Decision & 8s Escalation
6. **User / Contact** — Audible Siren & Care-giver Notification

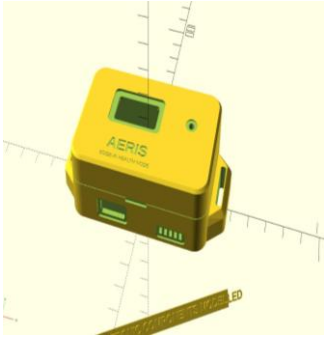
**OFFLINE PATHWAY:**
**(No Internet / Disasters)**

**LoRa Transceiver Module (433/868 MHz)**
Long-Range Offline Chirp Spread Spectrum Protocol

**Emergency / Remote Receiver (Gateway / Siren)**
Direct Offline Siren, Local Piezo Buzzer & First-Responder Alert



**PROPOSED HARDWARE ARCHITECTURE: CUSTOM PCB INTEGRATION**




**PROPOSED 3D CAD ENCLOSURE DESIGN**

# FEASIBILITY AND VIABILITY

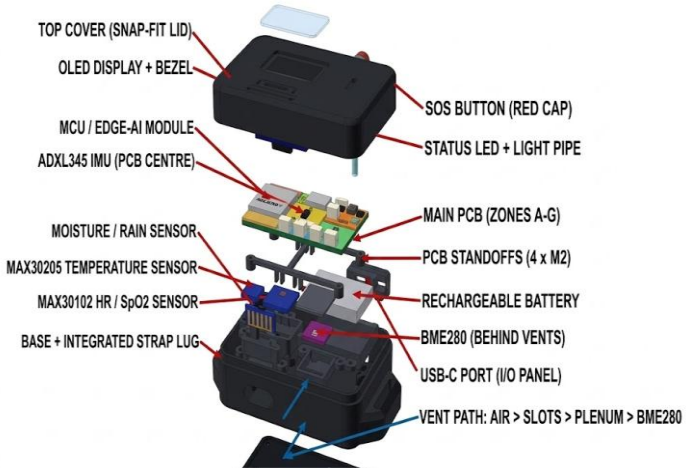
## Risk Index Becomes More Intelligent

## FEASIBILITY - Can we build it?

| AREA | AERIS HIGHLIGHT |
|---|---|
| 🧩 **Sensing** | Multi-modal health + motion + environment |
| 🧠 **Edge AI** | ESP32 + personal baseline |
| 📊 **Risk Engine** | Unified **0–100 Risk Index** |
| 🔧 **Custom PCB** | Proposed modular PCB design |
| 📐 **3D CAD** | Wearable enclosure **designed** |
| 📡 **Connectivity** | Internet + **LoRa** |
| 🧪 **Prototype** | Sensor acquisition validated on benchtop |

**AERIS: Intelligence in Every Layer**

• Compact • Intelligent
• Wearable • Deployment-Ready




| PRACTICAL CHALLENGE | PROPOSED MITIGATION |
|---|---|
| Sensor readings may be inaccurate | Multi-sensor cross-validation & auto-calibration |
| Internet connectivity unavailable | On-device edge processing & supported offline LoRa |
| False emergency alerts | Multi-tier thresholds & 8s confirmation/escalation |

## VIABILITY- Can it work in the real world?

| AREA | AERIS ADVANTAGE |
|---|---|
| 🔐 **Privacy** | Edge-first processing |
| 📡 **Resilience** | Online + offline LoRa |
| 🚨 **Safety** | Risk → Alert → Emergency |
| 🌦 **Context** | Health + environment + weather |
| 🧠 **Smart Stress** | GSR-free HRV-based estimation |
| 🔄 **Scalable** | Prototype → PCB → Wearable |
| 🏥 **Non-Diagnostic** | Risk awareness, not diagnosis |

# IMPACT AND BENEFITS

## BENEFITS




## IMPACT AT A GLANCE

| Impact Metric | AERIS Target / Value |
|---|---|
| Year-1 Reach Target | 10,000 pilot users (Target) |
| Estimated Prototype Cost | Estimated: ₹1,000–1,500 (BOM Component Target) |
| Processing Architecture | Edge-first, privacy-preserving on-device AI |
| Connectivity Resilience | Online (BLE/Wi-Fi) + Offline (LoRa) safety |
| Rapid Response Protocol | Risk Index → Buzzer → 8s confirmation → Alert |
| Long-Term Scalability Target | 1M+ users (Modular multi-form architecture target) |
| Risk Coverage Scope | Multi-modal: Vitals + Mobility + Environment + Weather |
| Market Growth Estimate | 11.49% CAGR (Preventive Healthcare Tech 2024–2032)* |

**ABDM** — Digital healthcare and interoperability

**NPCCHH** — Climate and environmental health

**Digital India / IndiaAI Mission-**
Secure, inclusive Edge-AI & IoT innovation

**Market Source: Research and Markets – Global Preventive Healthcare Technologies and Services Market Report (2024–2032), 11.49% CAGR.*

# RESEARCH AND REFERENCES

## RESEARCH

| Research Area | How it supports AERIS |
|---|---|
| Wearable Health Monitoring | HR, $SpO_2$ & temperature sensing |
| HRV & Stress Analysis | HRV-based personal risk patterns |
| Fall & Mobility Detection | Motion, gait & stability monitoring |
| Edge AI & Privacy | Local processing & reduced raw-data transmission |
| Environmental Health | Heat, humidity & weather-related risk |
| LoRa Emergency Communication | Offline long-range safety alerts |

## REFERENCES


### 🏛 NATIONAL POLICY & STANDARDS BENCHMARKS

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• **Digital Personal Data Protection (DPDP) Act, 2023:** Ministry of Law & Justice, Govt. of India (Aug 2023). Statutory compliance for on-device edge processing, user data minimization, and privacy-first architecture. *[Source: meity.gov.in / egazette.gov.in]*

• **National Programme on Climate Change & Human Health (NPCCHH):** NCDC & MoHFW, Govt. of India (2019–2024). Standard surveillance protocols for extreme heatwave indices and climate-sensitive physiological risk warnings. *[Source: ncdc.mohfw.gov.in]*

• **NDMA & Dept. of Telecommunications (DoT):** National Disaster Management Authority (2023–2024). Common Alerting Protocol (CAP) & Cell Broadcast emergency warning architecture. *[Source: ndma.gov.in]*

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• **Long-Range Low-Power Disaster Telemetry (LoRa):** Augustin, A., et al. (2016). 'A Study of LoRa: Long Range & Low Power Networks for IoT.' Sensors, 16(9), 1466. — *Validates sub-GHz offline emergency broadcast.*